

$$L_i = -[y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$

Starting at:

$$L_i = \log(1 + e^{\hat{y}_i}) - y_i \hat{y}_i$$

We can substitute $\hat{y}_i = \log(\frac{p_i}{1-p_i})$

$$= \log(1 + \frac{p_i}{1-p_i}) - y_i \log(\frac{p_i}{1-p_i})$$

Apply log rule of $\log(\frac{a}{b}) = \log a - \log b$ and $1 + \frac{p_i}{1-p_i} = \frac{1}{1-p_i}$

$$= \log(\frac{1}{1-p_i}) - y_i \log(\frac{p_i}{1-p_i}) = 0 - \log(1 - p_i) - y_i(\log(p_i) - \log(1 - p_i))$$

Take out the negative

$$= -[\log(1 - p_i) + y_i(\log(p_i) - \log(1 - p_i))]$$

Distribute y_i and rearrange

$$= -[y_i \log(p_i) + \log(1 - p_i) - y_i \log(1 - p_i)]$$

Factor out the $\log(1 - p_i)$ on the like terms

$$= -[y_i \log(p_i) + (1 - y_i) \log(1 - p_i)]$$